

**Syllabus:** Acoustics, Electrical, and Illumination Systems  
**Catalog Number:** ARC 4443  
**Instructor:** Kurt Neiswender, AIA  
**Office:** A 113  
**Office Hours:** [By Appointment](#)  
**Telephone:**  
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### Course Meetings:

**Lecture/Discussion:** Monday: 9:30 – 11:15 AM  
**Lecture/Discussion:** Wednesday: 9:30 – 10:45 AM  
**Room:** A223

### End-of-Semester Design Review Schedule

- Design reviews will be held May 5-7
- Specific time slots will be assigned to courses later in this semester.
- **Friday, May 2, Pencils Down!** No student work after 5:00 pm

Final presentations are part of a long tradition of sharing the work with outside critics and colleagues. The format simulates client presentations, pitches, and design reviews that occur in industry. Students must present their work as it is a requirement of a design studio. Students who do not present as part of the final review will not pass the course. Please make sure this is clear in both the syllabus and your course introduction. If your class is a lab and you are opting into the final review, this requirement can be adjusted.

### Covid-19 Protocol

LTU's Coronavirus policies and procedures can be found at <https://www.ltu.edu/coronavirus/>.

### Canvas: Online course management

The course syllabus, course calendar, assignments, additional readings and images, and other notes will be posted on the course Canvas site. Students are responsible for monitoring the site on a regular basis and for keeping up with coursework.

### Catalog Description

The course reviews the theories of sound perception, analysis of architectural acoustics, room acoustics, sound isolation, sound absorption, electronic sound reinforcement, and the study of acoustic properties of materials and room shapes to provide speech privacy, enhance speech clarity, enrich music spaces and control noise. The course also addresses basic electrical theory, utility connection options, emergency backup, distribution, and safety considerations as they relate to building construction. Electrical lighting is studied as part of an assimilated section covering the fundamentals of illumination, integrating natural daylighting and electrical lighting design for energy conservation. Class projects include investigations into course subjects to further enhance understanding of real work applications.

### Prerequisites

Integrated Design 2 - ARC 2126

## **Course Content**

This course is part of the building systems series and covers topics of lighting, electricity and acoustics systems integration in building design. The fundamental scientific principles governing light and sound in the built environment will be examined in the context of human physiological, psychological and biological needs as well as sustainable design perspective. It exposes students to technologies, materials and strategies for control of light and sound in buildings as well as the basic analyses needed to inform design decision-making and examine project performance. The course will continue the themes of resource efficiency and end-user comfort through the examination of emerging metrics for daylight sufficiency, visual, and acoustic comfort. The content is divided into key topics, progressing logically from foundational concepts to advanced systems, with practical applications and case studies. The knowledge gained in the systems courses are integral to your success in your Comprehensive Design Studio.

## **Required Text(s)**

Mechanical & Electrical Equipment for Buildings (MEEB) 13th Edition  
Walter T. Grondzik and Alison G. Kwok  
John Wiley and Sons, Inc. 2019

## **Recommended Text(s) and Resources**

Designing with Light: The Art, Science and Practice of Architectural Lighting Design,  
Jason Livingston  
Carbon-neutral Architectural Design 2nd Edition, Pablo La Roche  
The Coffee Sketch Podcast – [www.coffeesketchpodcast.com](http://www.coffeesketchpodcast.com)

## **NAAB Program Criteria (PC)**

Technical Knowledge: How the program ensures that students understand the established and emerging systems, technologies, and assemblies of building construction, and the methods and criteria architects use to assess those technologies against the design, economics, and performance objectives of projects.

### ***Environmental Systems: (SC.4B)***

Principles of environmental system design: Student is able to demonstrate understanding of basic principles of environmental systems design, based on qualitative and quantitative reasoning (I 01)

Application to problems: Student is able to demonstrate understanding of application of environmental systems principles and technologies to problems involving solar geometry, daylight, natural ventilation, heating and cooling, indoor air quality, lighting systems, and acoustics (R 02)

Geographic regions: Student is able to demonstrate understanding of using analysis to address environmental systems conditions across different geographic regions (E 03)

### ***Building Service Systems: (SC.4E)***

Principles of building service systems: Student is able to demonstrate understanding of basic principles of building service systems (including mechanical, plumbing, fire protection, electrical, and lighting systems) (I 01)

Application of building service systems: Student is able to demonstrate understanding of the appropriate application of building service systems (R 02)

Analysis of building service systems: Student is able to demonstrate understanding of performance analysis of building service systems through quantitative reasoning (R 03)

Application to problems: Student is able to demonstrate understanding of application of building service systems to practical and theoretical problems using advanced technologies (E 04)

## **Course Objectives**

The course work during this semester will emphasize building systems integration:

1. **Environmental Systems:** Ability to demonstrate the principles of environmental systems' design, how design criteria can vary by geographic region, and the tools used for performance assessment. This demonstration must include active and passive heating and cooling, solar geometry, daylighting, natural ventilation, indoor air quality, solar systems, lighting systems, and acoustics. (SC.4B)
2. **Building Service Systems:** Understanding of the basic principles and appropriate application and performance of building service systems, including lighting, mechanical, plumbing, electrical, communication, vertical transportation, security, and fire protection systems. (SC.4E)

## **Additional Learning Outcomes**

- **Ecological Knowledge:** Understanding of the dynamic between built and natural environments, of the impacts and the mitigation of climate change, and of the principles of resilience design.
- **Site Design:** Ability to respond to site characteristics, including urban context and developmental patterning, historical fabric, soil, topography, ecology, climate, and building orientation, in the development of a project design.
- **Health, Safety, and Welfare in the Built Environment:** How the program ensures that students understand the impact of the built environment on human health, safety, and welfare at multiple scales, from buildings to cities.
- **Building Integration:** How the program ensures that students develop the ability to make design decisions within architectural projects while demonstrating integration of building envelope systems and assemblies, structural systems, environmental control systems, life safety systems, and the measurable outcomes of building performance.

The course will be conducted in both lecture/seminar and lab formats which will involve theory, practical applications, calculations, and appropriate hands-on experiments with performance modeling tools. Topics are explained in lectures, practiced in the labs and integrated in the design process with the included assignments.

## **Common Course Assignments and Projects**

Course assignments are outlined below and will be posted to Canvas throughout the semester, and will be posted on Canvas. Each assignment is due on the designated day and time. Assignments will be listed on Canvas with the final due date and time. All completed assignments will be submitted digitally via Canvas. Do not email assignments as attachments. Due dates are tentative and may be adjusted in order to allow for coordination with students' studio work. Late assignment will be marked 5% per day (24-hr prorated) on grade. All assignments shall be graded for quality and content based on rubrics attached in Canvas. Sloppy, illegible, and disorganized works are not

acceptable for assignment and shall negatively impact your course grade.

### **Assignments:**

1. **Assignment #1:** Cove Introduction and basic input/output analysis. Daylighting Analysis
2. **Assignment #2:** Lighting Systems Design
3. **Assignment #3:** Electrical Systems Design
4. **Assignment #4:** Acoustical Design Optimization Project

### **Evaluation of Student Work**

Evaluation rubric will be posted on Canvas for each assignment of the project. Quizzes and exams will be automatically/manually evaluated and the results will be posted on Canvas. Quizzes and exam will be multiple choice, true or false, fill in the blank, and will include calculations and plots. Two tests will be given, midterm and final exam will be taken during the university assigned final exam week. No make-up test or exam, especially the final exam must be taken to pass the course.

### **Course Software**

**Cove:** [www.cove.tools](http://www.cove.tools)

“At Cove, we provide expert architectural and sustainability consulting that’s both cost-effective and efficient. Our AI-powered methods and advanced analytics accelerate projects without sacrificing quality. Trust us to deliver superior value and exceed your expectations with every project”.

To access LTU's licenses for Cove please go to the website above and create a new user account using your @ltu.edu email account. It should connect you to our account. If it does not please let me know as soon as possible. Many of you are already signed up, no new account work necessary.

Rhino/Grasshopper: pachyderm plug-in  
Revit: Lighting and Electrical Design

### **Course Grading Policy**

#### ***The grading policy is as follows:***

|                           |      |
|---------------------------|------|
| Attendance:               | 10%  |
| Discussion Participation: | 10%  |
| Quizzes:                  | 20%  |
| Assignments:              | 30%  |
| Exams:                    | 30%  |
|                           | 100% |
| Extra Credit:             | TBD  |

#### ***General grading rules:***

|           |             |
|-----------|-------------|
| <b>A</b>  | 94% - 100%  |
| <b>A-</b> | < 94% - 90% |
| <b>B+</b> | < 90% - 87% |
| <b>B</b>  | < 87% - 84% |
| <b>B-</b> | < 84% - 80% |
| <b>C+</b> | < 80% - 77% |
| <b>C</b>  | < 77% - 74% |
| <b>C-</b> | < 74% - 70% |
| <b>D+</b> | < 70% - 67% |
| <b>D</b>  | > 67% - 64% |
| <b>D-</b> | < 64% - 61% |
| <b>F</b>  | < 61%       |

### **Exception:**

The instructor reserves the right to adjust the low-end average cut-off score based upon the statistical distribution of the semester averages for final grades. For instance, if there is a large gap in the final

averages at 91%, then the instructor may decide to award an A grade to all students above that value. This can be done at the instructor's discretion and in no way should be construed to mean that it will be done each and every semester.

### Attendance:

Attendance will be checked at every class. The instructor holds the right to administratively drop (or to fail) the student if they exceed **three** absences. If you cannot come to class, contact your instructor before the class, via email. CoAD policy does not allow for any excused absences for any reason. If you arrive in the class later than 5 minutes, you will be marked as absent. If you leave the class during the lecture hours or lab session earlier than 5 minutes without prior notification, you will be marked as absent. For attendance credit calculation, one absence: no penalty > Two absences: -1 pts > Three absences: -3 pts > Four absences: -4 pts > Five absences: -5 pts (0 attendance pts) > Six absences: F grade (No exception by the University policy)

### The Student Companion

#### **Student Conduct, Academic Honesty, and Department Policies**

All of the college's policies and procedures have been compiled into a single document, The Student Companion. All students are responsible for reading and understanding these policies. The Companion is posted as a downloadable pdf document online for all faculty, staff, and students:  
[https://www.ltu.edu/architecture\\_and\\_design/student\\_resources.asp](https://www.ltu.edu/architecture_and_design/student_resources.asp)

#### **Topics covered in the Student Companion include:**

- The Studio Code
- Plagiarism
- Attendance and Late Work
- Participation

### Course Schedule

| Week          | Date                    | Topic                                   | Reading        | Quiz       | Assignments            | NAAB             |
|---------------|-------------------------|-----------------------------------------|----------------|------------|------------------------|------------------|
| <b>Part 1</b> |                         |                                         |                |            |                        |                  |
| Week 1        | January 13, 2025        | Design Process                          | Chapter 1      |            | Assignment #1 Released | SC4B-01, SC4B-03 |
|               | January 15, 2025        | Case Study: System Performance          |                |            |                        |                  |
| Week 2        | <b>January 20, 2025</b> | <b>NO CLASS</b>                         | <b>MLK Day</b> |            |                        | SC4B-01, SC4B-03 |
|               | January 22, 2025        | Daylighting, Solar Geometry             | Chapter 10     | Chapter 1  |                        |                  |
| <b>Part 2</b> |                         |                                         |                |            |                        |                  |
| Week 3        | January 27, 2025        | Lighting Fundamentals, Physics of Light | Chapter 6      |            |                        | SC4B-01, SC4B-03 |
|               | January 29, 2025        | Lighting Levels                         |                | Chapter 10 | Assignment #1 Due      |                  |
| Week 4        | February 3, 2025        | Electric Light Sources                  | Chapter 15     |            | Assignment #2 Released | SC4B-01, SC4B-03 |
|               | February 5, 2025        | Photometrics                            |                | Chapter 6  |                        |                  |
| Week 5        | February, 10, 2025      | Lighting Design Process                 | Chapter 16     |            |                        | SC4B-02, SC4E-01 |
|               | February 12, 2025       | Lighting Design Critiques               |                | Chapter 15 |                        |                  |
| Week 6        | February 17, 2025       | Electrical Lighting Design              | Chapter 17     |            |                        | SC4B-02, SC4E-01 |

|                    |                       |                                         |            |            |                        |                  |
|--------------------|-----------------------|-----------------------------------------|------------|------------|------------------------|------------------|
|                    | February 19, 2025     | Lighting Calculations                   |            | Chapter 16 |                        |                  |
| <b>Part 3</b>      |                       | <b>Electrical Systems</b>               |            |            |                        |                  |
| Week 7             | February 24, 2025     | Principles of Electricity               | Chapter 25 |            |                        | SC4B-02, SC4E-01 |
|                    | February 26, 2025     | Voltage, Power, Circuits                |            | Chapter 17 | Assignment #2 Due      |                  |
| Week 8             | March 3, 2025         | Electrical Systems and Materials        | Chapter 26 |            |                        | SC4B-02, SC4E-01 |
|                    | March 5, 2025         | Electrical Service Layout               |            | No quiz    |                        |                  |
| <b>Week 9</b>      | <b>March 10, 2025</b> | <b>SPRING BREAK</b>                     |            |            |                        |                  |
|                    | <b>March 12, 2025</b> | <b>SPRING BREAK</b>                     |            |            |                        |                  |
| Week 10            | March 17, 2025        | Electrical Wiring Design                | Chapter 28 |            |                        | SC4B-02, SC4E-01 |
|                    | March 19, 2025        | Wiring Diagrams                         |            | Chapter 26 |                        |                  |
| <b>Part 4</b>      |                       | <b>Acoustics</b>                        |            |            |                        |                  |
| Week 11            | March 24, 2025        | Fundamentals of Architectural Acoustics | Chapter 7  |            | Assignment #3 Released | SC4E-01, SC4E-02 |
|                    | March 26, 2025        | Sound Meter Exercise                    |            |            |                        |                  |
| Week 12            | March 31, 2025        | Sound in Enclosed Spaces                | Chapter 22 |            |                        | SC4E-01, SC4E-02 |
|                    | April 2, 2025         | Pachyderm                               |            | Chapter 28 | Assignment #3 Due      |                  |
| Week 13            | April 7, 2025         | Building Noise Control                  | Chapter 23 |            | Assignment #4 Released | SC4E-01, SC4E-02 |
|                    | April 9, 2025         | Case Study                              |            | Chapter 7  |                        |                  |
| <b>Part 5</b>      |                       | <b>Integration of Systems</b>           |            |            |                        |                  |
| Week 14            | April 14, 2025        | Integrated System Design                | Chapter 29 |            |                        | SC4E-03          |
|                    | April 16, 2025        | Comprehensive Design Goals              |            | Chapter 22 |                        |                  |
| Week 15            | April 21, 2025        | Successful Project Examples             |            |            |                        |                  |
|                    | April 23, 2025        | Case Study                              |            | Chapter 23 |                        |                  |
| Week 16            | April 28, 2025        | Comprehensive Design Critiques          |            |            |                        |                  |
|                    | April 30, 2025        |                                         |            | Chapter 29 | Assignment #4 Due      |                  |
| <b>Final Exams</b> | <b>May 7, 2025</b>    | <b>FINAL EXAM</b>                       |            |            |                        |                  |
| Grades Due         | May 14, 2025          |                                         |            |            |                        |                  |